

Cohomological and Categorical Foundations of Twin Operations: Deformation Theory, Rigidity, Quantum Groups, and Algebraic-Geometric Extensions

Jocelyn Nembé

Roots Insights Laboratory, Singapore
National Institute of Management Sciences, Libreville
jnembe777@gmail.com, jnembe@root-insights.org

Abstract

We develop the cohomological and categorical foundations of twin operations $x \odot_g y = g^{-1}(g(x) \alpha g(y))$, the conjugated operations central to the twin spaces framework. Our principal results are: (1) the **deformation cohomology** of twin operations is controlled by Hochschild cohomology — $\mathrm{HH}^1(G, \mathrm{End}(E))$ classifies infinitesimal deformations and $\mathrm{HH}^2(G, \mathrm{End}(E))$ contains obstructions to extension (Theorem 2); (2) a **rigidity dichotomy** for finite group actions: either $L = G$ (all elements preserve α , rigid) or $[G : L] \geq 2$ (flexible), by orbit-stabilizer; this orbit-flexibility is, however, *invisible* to $\mathrm{HH}^1(G, \mathrm{End}(E))$ in the semisimple case, where Maschke forces $\mathrm{HH}^1 = 0$ regardless of L , so cohomologically visible deformations require a non-semisimple setting (Theorem 5); (3) a structural **obstruction**: the quantum group $U_q(\mathfrak{sl}_2)$ is *not* a twin operation — the q -scaling is a Lie-algebra automorphism, so conjugation yields only the undeformed bracket, and a genuine quantum deformation needs a formal parameter rather than a group element (Remark 7); (4) twin operations form a **2-functor** from the groupoid of G -spaces to the 2-category of operation families (Theorem 9); (5) n -ary twin operations carry a natural **operad structure** compatible with Hochschild cohomology (Theorem 11).

We extend the framework to algebraic geometry: the kernel ideal $\mathfrak{k}_G = \{r \in R : g \cdot r = r\}$ of a twin ring (R, G) is the invariant subring R^G , connecting to the kernel concept of twin general relativity. Twin schemes are equivariant schemes in the sense of Mumford’s GIT, and the cohomological obstruction of the Cramer article lives in the Brauer group of the invariant ring.

Keywords: twin operations, Hochschild coho-

mology, deformation theory, rigidity, quantum groups, operads, equivariant geometry

MSC 2020: 20M30, 18M60, 18G60, 16E40, 17B37

1 Introduction

The twin operations $x \odot_g y = g^{-1}(g(x) \alpha g(y))$, introduced in the foundational theory of twin spaces [9, 10], appear throughout the ROOTS-INSIGHTS program: in information theory (orbit entropy [11]), in relativity (Einstein addition as $\odot_g$ with $g = \mathrm{artanh}$ [12]), and in linear algebra (Cramer’s rule with cohomological obstruction in $H^2(G, K_0^\times)$ [13]).

This article develops the *algebraic foundations*: what governs the space of twin operations, when are they rigid, how they relate to quantum groups, and what categorical and operadic structures they carry. We also establish the connection to equivariant algebraic geometry, showing that the twin framework provides a unified language that encompasses Mumford’s geometric invariant theory [6].

2 Deformation Cohomology of Twin Operations

Definition 1 (Hochschild Complex for Twin Operations). Let (E, α, G) be a magma with group action. The Hochschild cochain complex is $C^m(G, \mathrm{End}(E)) = \{f : G^m \rightarrow \mathrm{End}(E)\}$ with dif-

ferential:

$$\begin{aligned}
(\delta^n f)(g_1, \dots, g_{n+1}) &= g_1 \cdot f(g_2, \dots, g_{n+1}) \\
&+ \sum_{i=1}^n (-1)^i f(g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) \\
&+ (-1)^{n+1} f(g_1, \dots, g_n)
\end{aligned}$$

where $(g \cdot \phi)(x) = g \cdot \phi(g^{-1} \cdot x)$ is the conjugation action. The cohomology groups are $\mathrm{HH}^n(G, \mathrm{End}(E)) = \ker \delta^n / \mathrm{im} \delta^{n-1}$.

Theorem 2 (Deformation Cohomology). *Let (E, α, G) be a magma with group action. Then:*

- (a) $\mathrm{HH}^0(G, \mathrm{End}(E)) = \mathrm{End}(E)^G$ is the ring of G -invariant endomorphisms.
- (b) $\mathrm{HH}^1(G, \mathrm{End}(E))$ classifies **infinitesimal deformations** of α through G : an element $[\eta] \in \mathrm{HH}^1$ corresponds to a first-order deformation $\alpha + \varepsilon\eta + O(\varepsilon^2)$ compatible with the G -action.
- (c) $\mathrm{HH}^2(G, \mathrm{End}(E))$ contains **obstructions**: given a first-order deformation η , the obstruction to extending it to second order is a class $[\omega(\eta)] \in \mathrm{HH}^2$. The deformation extends iff $[\omega(\eta)] = 0$.
- (d) The **linear subgroup** $L = \{g \in G : \odot_g = \alpha\} = G \cap \mathrm{Aut}(E, \alpha)$ is the stabilizer of α under the transport action $g \mapsto \odot_g$; hence $\mathcal{O} \cong L \backslash G$ and $|\mathcal{O}| = [G : L]$. It is a subgroup of G (not a cohomology group), and $L = G$ characterizes the rigid case of §3.

Proof. This follows the classical Gerstenhaber deformation theory [1] applied to the specific setting of conjugated operations.

(a) $\mathrm{HH}^0 = \ker \delta^0 = \{\phi \in \mathrm{End}(E) : g \cdot \phi = \phi \forall g\} = \mathrm{End}(E)^G$.

(b) An infinitesimal deformation of α compatible with G is a family $\alpha_\varepsilon(x, y) = \alpha(x, y) + \varepsilon\eta(x, y) + O(\varepsilon^2)$ such that $g^{-1}(\alpha_\varepsilon(g \cdot x, g \cdot y))$ varies smoothly with g . The compatibility condition is $\delta^1 \eta = 0$, and two deformations are equivalent iff they differ by $\delta^0 \phi$ for some $\phi \in C^0$. So infinitesimal deformations $= \ker \delta^1 / \mathrm{im} \delta^0 = \mathrm{HH}^1$.

(c) Given η with $\delta^1 \eta = 0$, the second-order extension $\alpha + \varepsilon\eta + \varepsilon^2 \mu$ requires $\delta^1 \mu = -\frac{1}{2}[\eta, \eta]$ where $[\eta, \eta]$ is the Gerstenhaber bracket. The class $[\omega(\eta)] = [\frac{1}{2}[\eta, \eta]] \in \mathrm{HH}^2$ is the obstruction. If $[\omega] = 0$: $\exists \mu$ solving the equation.

(d) $g \in L$ iff $\odot_g = \alpha$ iff $g^{-1}(\alpha(g \cdot x, g \cdot y)) = \alpha(x, y)$ for all x, y , i.e. $\alpha(g \cdot x, g \cdot y) = g \cdot \alpha(x, y)$: exactly the condition that the bijection g be an automorphism of (E, α) . Hence $L = G \cap \mathrm{Aut}(E, \alpha)$, the stabilizer of α , and $|\mathcal{O}| = [G : L]$ by orbit-stabilizer. $\square$

Corollary 3 (Connection to Cramer Obstruction). *The cohomological obstruction $[c_P] \in H^2(G, K_0^\times)$ of the Cramer article [13] is a special case: the coefficient module is $K_0^\times \subset \mathrm{End}(K_0)$, and $[c_P] = 0$ for Cramer's rule because the determinant is a natural transformation (hence $\eta = 0$, no deformation needed).*

3 Rigidity Dichotomy

Definition 4 (Orbit of Operations). The **orbit of α under G** is $\mathcal{O} = \{\odot_g : g \in G\} \cong L \backslash G$ (the space of right cosets, via $Lg \mapsto \odot_g$), where $L = \{g : \odot_g = \alpha\}$ is the linear subgroup, the stabilizer of α under the right transport action $g \mapsto \odot_g$. In particular $|\mathcal{O}| = [G : L]$.

Theorem 5 (Rigidity Dichotomy). *Let (E, α, G) with G finite. Exactly one of:*

- (a) **RIGID**: $\mathcal{O} = \{\alpha\}$, i.e., $L = G$. All elements of G preserve α .
- (b) **FLEXIBLE**: $|\mathcal{O}| \geq 2$, i.e., $[G : L] \geq 2$. There exist genuinely distinct twin operations.

This orbit dichotomy is governed entirely by orbit-stabilizer; rigidity means $L = G$, i.e. $G \subseteq \mathrm{Aut}(E, \alpha)$.

Relation to cohomology. $\mathrm{HH}^1(G, \mathrm{End}(E))$ measures infinitesimal deformations (Theorem 2b), a strictly finer notion than orbit-flexibility. For finite G with $|G|$ invertible in K , Maschke's theorem forces $\mathrm{HH}^n(G, \mathrm{End}(E)) = 0$ for all $n \geq 1$ regardless of L ; so in the semisimple case $\mathrm{HH}^1 = 0$ always and does not detect flexibility (see Example 6, which is flexible yet has $\mathrm{HH}^1 = 0$). Cohomologically visible deformations require a non-semisimple setting (modular characteristic, or positive-dimensional/Lie groups). For abelian G :

- If α is a group operation and $G \subseteq \mathrm{Aut}(E, \alpha)$: **RIGID**.
- If $G \not\subseteq \mathrm{Aut}(E, \alpha)$: **FLEXIBLE** (and $[G : L]$ divides $|G|$).

Proof. The dichotomy is the orbit-stabilizer theorem: $|G| = |L| \cdot |\mathcal{O}|$, so $|\mathcal{O}| = 1$ or $|\mathcal{O}| \geq 2$.

Cohomology in the semisimple case. If G is finite and $\mathrm{char}(K) \nmid |G|$, then KG is semisimple (Maschke), every module is projective, and $\mathrm{HH}^n(G, \mathrm{End}(E)) = H^n(G, \mathrm{End}(E)) = 0$ for all $n \geq 1$. This holds whether or not $L = G$; in particular $\mathrm{HH}^1 = 0$ does *not* imply rigidity. The cohomological vanishing is automatic here, independent of the orbit dichotomy.

For abelian G : if $G \subseteq \mathrm{Aut}(E, \alpha)$, then every $g \in G$ is an automorphism of (E, α) , so $\odot_g =$

$g^{-1} \circ \alpha \circ (g \times g) = \alpha$. Thus $L = G$ (rigid). If some $g \notin \text{Aut}(E, \alpha)$, then $\odot_g \neq \alpha$, so $L \subsetneq G$ and $[G : L] \geq 2$ (flexible). $\square$

Example 6 (Rigid vs. Flexible). Let $E = \mathbb{R}$, $\alpha = +$. **(a) Rigid.** $G = \{\text{id}, s\} \cong \mathbb{Z}/2$ with $s(x) = -x$. Since s is an automorphism of $(\mathbb{R}, +)$, $\odot_s(x, y) = -(-x + (-y)) = x + y$, so $L = G$: RIGID. **(b) Flexible.** $G = \{\text{id}, r\} \cong \mathbb{Z}/2$ with the reflection $r(x) = c - x$, $c \neq 0$. Here r is *not* additive, and $\odot_r(x, y) = r(r(x) + r(y)) = r(2c - x - y) = x + y - c \neq +$, so $L = \{\text{id}\}$ and $[G : L] = 2$: FLEXIBLE. Both cases have $\text{HH}^1(G, \text{End}(\mathbb{R})) = 0$ (Maschke), so flexibility in (b) is invisible to HH^1 — illustrating the decoupling above.

4 Quantum Groups: Analogy, Not Realization

Remark 7 (Quantum groups are *not* twin operations — only an analogy). It is tempting to read the quantum group $U_q(\mathfrak{sl}_2)$ ($q \in \mathbb{C}^*$) as a twin operation, with the parameter q playing the role of the group element and $G = (\mathbb{C}^*, \cdot)$ acting on $\mathfrak{sl}_2$ by the q -scaling $\phi_q : H \mapsto H$, $E \mapsto q^{1/2}E$, $F \mapsto q^{-1/2}F$. This analogy *fails as a literal realization*, for a structural reason. The map ϕ_q is a *Lie-algebra automorphism* of $\mathfrak{sl}_2$: indeed $[q^{1/2}E, q^{-1/2}F] = [E, F] = H$, $[H, q^{1/2}E] = 2q^{1/2}E$, and $[H, q^{-1/2}F] = -2q^{-1/2}F$. Hence every q lies in the linear subgroup L , and the twin bracket is the *undeformed* one,

$$[X, Y]_q := \phi_q^{-1}([\phi_q X, \phi_q Y]) = [X, Y] \quad \text{for all } q,$$

so in particular $[E, F]_q = H$ (not $[H]_q$): this is the *rigid* case of §3. By contrast the genuine Jimbo–Drinfeld relation is

$$[E, F] = \frac{K - K^{-1}}{q - q^{-1}}, \quad KEK^{-1} = q^2 E, \quad KFK^{-1} = q^{-2} F,$$

whose right-hand side involves a *new* generator $K = q^H$ rather than a fixed linear expression in H ; as $q \rightarrow 1$ it degenerates to $[E, F] = H$, but no conjugation ϕ_q can manufacture the K -dependence, which confirms that the deformation is genuine and lies outside the conjugation orbit. More fundamentally, any twin operation $\odot_g$ is isomorphic to its base α via g (Theorem 2(d)), so conjugation can never yield a genuine quantum deformation — $U_q(\mathfrak{sl}_2)$ is not isomorphic to $U(\mathfrak{sl}_2)$ as a Hopf algebra. A faithful q -deformation requires a *formal* parameter (Drinfeld twist / $\hbar$ -deformation;

cf. Drinfeld [3], Jimbo [4], and Remark 8), which is genuinely different from—though formally reminiscent of—the rigid q -scaling. The twin framework thus *locates* quantization as a phenomenon lying outside the conjugation orbit, rather than realizing it.

Remark 8 (Connection to Kontsevich formality). The Kontsevich formality theorem [2] states that every Poisson manifold admits a formal deformation quantization. In twin language: the deformation space $\mathcal{O}$ of the Poisson bracket under the group of formal diffeomorphisms is non-empty (FLEXIBLE), and the formality map identifies HH^2 with the space of Poisson bivectors.

5 Categorical Structure: Twin Operations as 2-Functor

Theorem 9 (2-Functor Structure). *The assignment:*

$$\text{Objects: } (E, \alpha, G) \mapsto \{\odot_g : g \in G\} \quad (1)$$

$$1\text{-morphisms: } \phi : (E_1, \alpha_1, G) \rightarrow (E_2, \alpha_2, G) \mapsto \{\phi_g\} \quad (2)$$

$$2\text{-morphisms: } \text{natural transformations between families} \quad (3)$$

defines a 2-functor $\text{Twin} : \mathbf{G-Mag} \rightarrow \mathbf{OpFam}$ *from the 2-category of G -magmas to the 2-category of operation families.*

Proof sketch. Functoriality on 1-morphisms: if $\phi : (E_1, \alpha_1) \rightarrow (E_2, \alpha_2)$ is a G -equivariant magma homomorphism, then $\phi_g : (E_1, \odot_g^{(1)}) \rightarrow (E_2, \odot_g^{(2)})$ is a homomorphism for each g . Compatibility with composition follows from $\phi \circ \psi$ being G -equivariant iff both ϕ and ψ are. The 2-morphisms (natural transformations) encode deformations of morphisms compatible with the G -action. $\square$

6 Operadic Structure of n -ary Twin Operations

Definition 10 (n -ary Twin Operation). For an n -ary operation $\alpha : E^n \rightarrow E$ and $g \in G$: $\alpha_g(x_1, \dots, x_n) = g^{-1}(\alpha(g \cdot x_1, \dots, g \cdot x_n))$.

Theorem 11 (Operad Structure). *The collection* $\mathcal{P}(n) = \{\alpha_g^{(n)} : g \in G\}$ *of n -ary twin operations carries the structure of an operad:*

(a) **Composition:** $\gamma : \mathcal{P}(n) \times \mathcal{P}(k_1) \times \dots \times \mathcal{P}(k_n) \rightarrow \mathcal{P}(k_1 + \dots + k_n)$ *by substitution.*

- (b) **Equivariance:** S_n acts on $\mathcal{P}(n)$ by permuting inputs; this action commutes with the G -transport.
- (c) **Unit:** $\text{id} \in \mathcal{P}(1)$ is the identity.
- (d) **Cohomological compatibility:** The operadic cohomology of $\mathcal{P}$ is related to Hochschild cohomology by a spectral sequence: $E_2^{p,q} = H^p(\mathcal{P}, \text{HH}^q(G, \text{End}(E))) \Rightarrow H_{\text{op}}^{p+q}(\mathcal{P})$.

Proof sketch. (a) Substitution: $(\alpha_g \circ (\beta_{h_1}, \dots, \beta_{h_n}))(x_{11}, \dots, x_{nk_n}) = g^{-1}(\alpha(g \cdot \beta_{h_1}(\dots), \dots, g \cdot \beta_{h_n}(\dots)))$. This is well-defined and associative by associativity of function composition. (b) S_n permutes the inputs: $(\sigma \cdot \alpha_g)(x_1, \dots, x_n) = \alpha_g(x_{\sigma(1)}, \dots, x_{\sigma(n)})$. Since g acts on each input independently, the S_n -action commutes with G -transport. (c) The identity $\text{id}(x) = x$ satisfies $\text{id}_g = g^{-1} \circ g = \text{id}$. (d) The spectral sequence arises from filtering the operadic cochain complex by the degree in G . $\square$

Remark 12 (Connection to Loday–Vallette). The operad $\mathcal{P}$ is a special case of the Loday–Vallette theory of algebraic operads [5]: the twin operations form a *parameterized operad* (an operad fibered over the group G). The Koszul duality of $\mathcal{P}$ encodes the relation between associative and Lie twin operations.

7 Classification for Finite Groups

Theorem 13 (Classification). *For finite group G acting on a finite set E with $|E| = n$:*

- (a) *The number of distinct twin operations is $|\mathcal{O}| = [G : L]$, dividing $|G|$.*
- (b) *For $G = \mathbb{Z}/m\mathbb{Z}$ cyclic: $|\mathcal{O}|$ divides m , and the orbit of α under G is periodic with period $[G : L]$.*
- (c) *For $G = S_n$ symmetric: the orbit $\mathcal{O}$ is determined by the cycle type of the permutations in L . The number of orbits is related to the number of conjugacy classes of subgroups of S_n .*
- (d) *For α associative: $L \supseteq \text{Aut}(E, \alpha) \cap G$ (all group automorphisms that are also in G preserve α). If $G \subseteq \text{Aut}(E, \alpha)$: RIGID.*

Proof. (a) Orbit-stabilizer theorem. (b) For $\mathbb{Z}/m$: L is a subgroup, hence $|L|$ divides m , so $|\mathcal{O}| = m/|L|$ divides m . Periodicity: $\odot_{g^k} = \odot_{g^{k+[G:L]}}$. (c) For S_n : L is determined by which permutations commute with α in the required sense. The conjugacy class structure of S_n partitions the permutations into families with the same orbit type. (d) If $g \in \text{Aut}(E, \alpha)$: $g^{-1}(\alpha(g \cdot x, g \cdot y)) = g^{-1}(g(\alpha(x, y))) = \alpha(x, y)$, so $g \in L$. $\square$

Example 14 (Three twin additions on $\mathbb{Z}/3$). Let $E = \mathbb{Z}/3$, $\alpha = + \bmod 3$, and $G = S_3 = \text{Sym}(E)$ in its defining action. By Theorem 2(d), $L = G \cap \text{Aut}(\mathbb{Z}/3, +)$. An additive automorphism sends the generator to a generator, so $\text{Aut}(\mathbb{Z}/3, +) = \{\text{id}, x \mapsto -x\}$ has order 2 (the map $x \mapsto -x$ is the transposition (1 2), fixing 0). Hence $|L| = 2$ and $[G : L] = 6/2 = 3$: there are exactly three distinct twin operations, one per right coset of L , all isomorphic to $+$ but with distinct Cayley tables. This is a fully explicit FLEXIBLE case; yet, being a finite group in characteristic 0, it has $\text{HH}^1 = 0$, illustrating once more that orbit-flexibility is strictly finer than the cohomological flexibility measured by HH^1 .

8 Algebraic-Geometric Extension: Twin Rings and Schemes

We now extend the twin framework to commutative algebra and algebraic geometry, connecting to equivariant geometry [6, 7].

Definition 15 (Twin Ring). A **twin ring** is a pair (R, G) where R is a commutative ring and G acts on R by ring automorphisms. The **kernel ideal** is:

$$\mathfrak{k}_G = R^G = \{r \in R : g \cdot r = r \ \forall g \in G\} \quad (4)$$

This is the **invariant subring** (a subring, not an ideal — the name “kernel ideal” is by analogy with the kernel of Vol. 17 [12]).

Remark 16 (Connection to equivariant geometry). A twin ring (R, G) is precisely a G -ring in the sense of equivariant algebra [6]. The twin scheme $\text{Spec}(R, G)$ is the G -scheme $\text{Spec}(R)$ with the induced G -action. We use “twin” terminology to maintain consistency with the ROOTS-INSIGHTS framework, but acknowledge that the mathematical content is Mumford’s geometric invariant theory applied to the twin setting.

Proposition 17 (Structure of Twin Rings). *For a Noetherian twin ring (R, G) with G finite, $|G|$ invertible in R :*

- (a) R^G is Noetherian (Hilbert’s theorem).
- (b) R is integral over R^G (lying-over theorem).
- (c) The quotient map $\pi : \text{Spec}(R) \rightarrow \text{Spec}(R^G)$ is the orbit map: points of $\text{Spec}(R^G)$ correspond to G -orbits of prime ideals.
- (d) The Brauer group $\text{Br}(R^G)$ controls obstructions to splitting twin modules, connecting to the $H^2(G, K_0^\times)$ obstruction of the Cramer article.

Proof sketch. (a) By Noether’s theorem (E. Noether, 1926 [8]): if R is Noetherian and G is finite, R^G is Noetherian. (b) For $r \in R$: the polynomial $\prod_{g \in G} (x - g \cdot r) \in R^G[x]$ has r as a root, so r is integral over R^G . (c) Standard GIT: $\text{Spec}(R^G) = \text{Spec}(R)//G$. (d) The Brauer group $\text{Br}(R^G)$ classifies Azumaya algebras over R^G ; the image of $H^2(G, R^\times)$ in $\text{Br}(R^G)$ measures obstructions to descending R -modules to R^G -modules. $\square$

Example 18 (CM Elliptic Curves). An elliptic curve $E/\mathbb{Q}$ with complex multiplication by $\mathcal{O}_K$ (ring of integers of an imaginary quadratic field K) is a twin variety (E, G) with $G = \mathcal{O}_K^\times$. The Tate module $T_\ell(E)$ carries commuting actions of $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ and G . The kernel ideal R^G encodes the CM invariants, and the Galois representation factors through G , leading to explicit class field theory.

9 Conclusion

The twin operations $\odot_g$ carry rich cohomological structure (Hochschild cohomology controlling deformations), categorical structure (2-functor), and operadic structure (n -ary extension). The rigidity dichotomy provides a clean criterion for when the twin hierarchy is trivial vs. non-trivial. Quantum groups, by contrast, do *not* arise from the twin (conjugation) construction: the q -scaling is an automorphism, hence rigid, so genuine quantization requires a formal deformation parameter rather than a group element (Remark 7).

The extension to algebraic geometry shows that twin rings and schemes are equivariant objects in the sense of Mumford’s GIT, with the kernel ideal R^G playing the same role as the isometry group $\text{Ker}(g)$ in twin GR. The Brauer group obstruction connects the cohomological theory to the Cramer article’s H^2 -obstruction.

Open problems: (1) Koszul duality for the twin operad and its relation to Lie-type twin operations. (2) Derived categories of twin sheaves and connection to motivic cohomology. (3) Higher categorical extensions (twin ∞ -operads). (4) The twin Langlands correspondence: whether the functorial transport extends to automorphic forms.

References

- [1] M. Gerstenhaber, “On the deformation of rings and algebras,” *Ann. Math.*, vol. 79, pp. 59–103, 1964.
- [2] M. Kontsevich, “Deformation quantization of Poisson manifolds,” *Lett. Math. Phys.*, vol. 66, pp. 157–216, 2003 (preprint 1997).
- [3] V.G. Drinfeld, “Quantum groups,” in *Proc. ICM Berkeley*, AMS, pp. 798–820, 1987.
- [4] M. Jimbo, “A q -difference analogue of $U(\mathfrak{g})$,” *Lett. Math. Phys.*, vol. 10, pp. 63–69, 1985.
- [5] J.-L. Loday, B. Vallette, *Algebraic Operads*, Springer, 2012.
- [6] D. Mumford, *Geometric Invariant Theory*, Springer, 1965.
- [7] R.W. Thomason, “Algebraic K -theory of group scheme actions,” *Ann. Math. Studies*, vol. 113, pp. 539–563, 1987.
- [8] E. Noether, “Der Endlichkeitssatz der Invarianten endlicher Gruppen,” *Math. Ann.*, vol. 77, pp. 89–92, 1926.
- [9] J. Nembé, “Twin Spaces: Foundations,” *ROOTS-INSIGHTS*, Vol. 1, 2026.
- [10] J. Nembé, “The Transfer Theorem,” *ROOTS-INSIGHTS*, Vol. 3, 2026.
- [11] J. Nembé, “Twin Information Theory,” *ROOTS-INSIGHTS*, Vol. 16, 2026.
- [12] J. Nembé, “Twin General Relativity,” *ROOTS-INSIGHTS*, Vol. 17, 2026.
- [13] J. Nembé, “Cramer’s Rule in Twin Spaces,” 2026.